

Elizabeth McDaniel, PhD

Scientist with a passion for bioinformatics, developing reproducible tools that scale, and helping researchers with their analytical challenges.

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Skills

Programming: Python, R, bash, unix, Markdown, SQL (MySQL)

Workflow Management and Infrastructure: Nextflow, Docker, conda, git/GitHub, GitHub Actions for CI/CD, AWS (Batch, EC2, S3, ECR), SLURM, HT-Condor, Jupyter Lab, static site development (GitHub pages, Jekyll, Quarto), Linux systems

Bioinformatics: Genome/metagenome assembly (short and long-read), genome annotation, bulk RNA-seq analysis, amplicon sequencing, variant calling, protein structure analysis, machine learning applications for biological data (clustering, dimensionality reduction, classification)

Wet-lab: DNA/RNA extraction, PCR, qPCR, HPLC, lab-scale bioreactor operation

Communications and Other Skills: Data science workshop development and instructor (Certified Carpentries instructor since 2019), Adobe Illustrator, Microsoft Office Suite, manuscript writing

Education

University of Wisconsin - Madison

2016-2021

Ph.D., Microbiology

University of Arkansas

2012-2016

B.S., Biology with Minor in Statistics

Professional and Research Experience

Computational Biologist – *Astera Institute/Microcosm Foods*

Sep. 2024-present

- Created reproducible computational workflows in Nextflow for predicting bioactive peptides, detecting microbial species in publicly available metagenomic samples, and annotating thousands of bacterial genomes with multiple programs.
- Led external outreach and open science strategy, depositing data in FAIR repositories, and working with external partners for extending the reach and reuse of public datasets.
- Incorporated metagenomic analyses to select appropriate commercial starter culture strains for developing a novel fermented food product prototype and conducted customer discovery interviews to assess product-market fit.

Computational Scientist - *Arcadia Science*

Oct. 2022-Aug 2024

- Created reproducible computational workflows in Nextflow for both bespoke analyses such as RNA sequence mapping and protein structure analysis and production-ready pipelines for metagenomics and genome assembly on AWS Batch, AWS EC2 instances, and/or Nextflow Tower.
- Assisted scientists with their analytical challenges for pilot projects on quick timescales, provided pre-publication code review, and created internal tools to help scientists understand and visualize their data.
- Communicated the development of open-source tools and scientific results through a non-journal-based publishing platform.

Postdoctoral Researcher - *University of British Columbia, Vancouver, BC*

Nov. 2021-Sep. 2022

- Explored the composition and metabolic activity of microbial communities underlying anaerobic digestion systems using genome-resolved metagenomics, metaproteomics, and stable-isotope probing techniques.

Graduate Research Assistant – *University of Wisconsin – Madison, Madison, WI*

Aug. 2016-Aug. 2021

- Dissertation on the ecology and evolution of enrichment microbial communities simulating enhanced biological phosphorus removal using genome-resolved metagenomics and metatranscriptomics.
- Assisted other researchers with high-throughput mining and annotation of microbial metagenomes.
- Taught workshops on reproducible coding skills with R, python, git/GitHub as a Certified Carpentries Instructor.

Selected Publications

Discovering shared protein structure signatures connected to polyphosphate accumulation in diverse bacteria. Avasthi P., Celebi F.M., McDaniel E.A. *Arcadia Science*. Oct. 2023. DOI: 10.57844/arcadia-ac10-23e7

Diverse electron carriers drive syntrophic interactions in an enriched anaerobic acetate-oxidizing consortium. McDaniel E.A., Scarborough M., Mulat G.M., Lin X., Sampara P.S., Olson H.M., Young R.P., Eder E.K., Attah I.K., Markillie L.M., Hoyt D.W., Lipton M.S., Hallam S.J., Ziels R.M. *ISMEJ*. Oct. 2023. DOI: 10.1038/s41396-023-01542-6.

Quickly preprocessing and profiling microbial community sequencing data with a Nextflow workflow for metagenomics. Dutton R.J., McDaniel E.A. *Arcadia Science*. May 2023. DOI: 10.57844/arcadia-7etp-pj24

TbasCO: Trait-based Comparative 'Omics Identifies Ecosystem-Level and Niche-Differentiating Adaptations of an Engineered Microbiome. McDaniel E.A.#, van Steenbrugge J.J.M.#, Noguera D.R., McMahon K.D., Raaijmakers J.M., Medema M.H., Oyserman B.O. *bioRxiv*. Dec. 2021. DOI: 10.1101/2021.12.04.471239

Expanded Phylogenetic Diversity and Metabolic Flexibility of Mercury-Methylating Microorganisms. McDaniel E.A., Peterson B., Stevens S.L.R., Tran P.Q., Anantharaman K., McMahon K.D. *mSystems*. Aug. 2020. DOI: 10.1128/mSystems.00299-20